

PhD in Machine Learning specialized in interpretability to build safe, reliable, and trustworthy AI for critical applications.

Education

- 2025 **Technical University Berlin**, *Dr. rer. nat. (summa cum laude)*, Advisor: Prof. Dr. Wojciech Samek, Title: “*Interpretability-Driven Model Debugging with the Reveal2Revise Life Cycle*”.
- 2018 **Humboldt University Berlin**, *M.Sc. in Information Systems*, Grade: 1.2 (very good), top 10%.
- 2017 **University of Oslo**, *Semester abroad*, Master level courses.
- 2015 **DHBW Mannheim**, *B.Sc. in Information Systems*, Grade: 1.4 (very good), top 10%.

Work Experience

- 2021 – today **Research Associate**, *Fraunhofer Heinrich-Hertz Institute*, Berlin (Germany).
PhD student and postdoctoral researcher in the eXplainable AI group led by Dr. Lapuschkin and Prof. Dr. Samek, focusing on interpretability-driven model debugging. Development and integration of XAI methods into the AI development life cycle to verify, validate, and improve model behavior, ultimately increasing the reliability and trustworthiness of AI systems, particularly in medical applications.
- 2019 – 2021 **Machine Learning Scientist**, *Amazon.com, Inc.*, Edinburgh (United Kingdom).
Applied machine learning scientist at Amazon, focusing on explainability and fairness in ML for recruiting technologies.
- 2017 – 2019 **Master’s Thesis Student / Research Associate**, *SAP AI Research*, Berlin (Germany).
Master’s thesis student and full-time research associate with a focus on multimodal few-shot learning.
- 2017 **Machine Learning Developer (Intern)**, *Bakken & Bæck*, Oslo (Norway).
Text embedding learning, language detection, data retrieval, data preparation, and prototype development.
- 2015 – 2017 **Working Student**, *SAP SE - Medical Research Insights*, Potsdam (Germany).
Infrastructural topics, product development and ML prototypes for medical research insights.
- 2012 – 2015 **Bachelor Student (Cooperative Education)**, *SAP SE*, Walldorf (Germany).
During my bachelor’s program I was employed by SAP SE and have worked in several development teams.

Selected Publications (see Google Scholar for Full List)

- 2025 **Ensuring Medical AI Safety: Interpretability-Driven Detection and Mitigation of Spurious Model Behavior and Associated Data**, F. Pahde, T. Wiegand, S. Lapuschkin, W. Samek, *Machine Learning — Special Issue on XAI for Secure Applications*.
- 2025 **Navigating Neural Space: Revisiting Concept Activation Vectors to Overcome Directional Divergence**, F. Pahde, M. Dreyer, L. Weber, M. Weckbecker, C. Anders, T. Wiegand, W. Samek, S. Lapuschkin, *ICLR 2025*.
- 2024 **From Hope to Safety: Unlearning Biases of Deep Models via Gradient Penalization in Latent Space**, M. Dreyer*, F. Pahde*, C. Anders, W. Samek, S. Lapuschkin, *AAAI 2024 (oral)*.
- 2023 **Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models**, F. Pahde*, M. Dreyer*, W. Samek, S. Lapuschkin, *MICCAI 2023*.
- 2023 **Optimizing Explanations by Network Canonization and Hyperparameter Search**, F. Pahde, G. Yolcu, A. Binder, W. Samek, S. Lapuschkin, *CVPRW 2023*.
- 2021 **Multimodal Prototypical Networks for Few-shot Learning**, F. Pahde, M. Puszcz, T. Klein, M. Nabi, *WACV 2021*.

Computer Skills

Programming Python, Tensorflow, Keras, PyTorch, PyTorch-Lightning, zennit, scikit-learn, Java, R, Octave, MySQL
Other Tools Git, L^AT_EX, Microsoft Office

Languages

German
English

Native
Fully Professional